1.	**Name of Course :**	Engineers and Society				
	Course Code :	EEE202				
2.	**Synopsis :**	This course aims to explain the main concepts in engineering ethics, risk management and occupational safety and health as well as to expose the students to basic of law in the engineering context. The materials will be of introductory nature to enable engineers to appreciate factors that have to be taken into account in decision-making. At the end of the course, students will be able to identify and discuss issues and challenges faced by engineers relating to engineering ethics, risk management and to understand the legal requirements related to engineering field.				
3.	**Name(s) of academic staff :**	Ir. Dr Liew Chia Pao				
4.	**Semester and Year offered :**	**Semester**	1	**Year**	2	
5.	**Credit Value :**	3				
6.	**Prerequisite/ co-requisite: (if any)**	None				

7.	**Course Learning Outcomes (CLO) : At the end of the course the students will be able to: (example) - explain the basic principles of immunisation (C2,PLO1)**		**Bloom's Taxonomy Level**	**PLO**
	CLO1	Apply principles of ethics of a professional engineer in relation to society and norms of engineering practice.	3	8
	CLO2	Analyse the impact of the technological developments with appropriate consideration for societal, health, safety and legal factors, and the consequent responsibilities relevant to professional engineering practices.	4	6
	CLO3	Analyse the key issues in the use of natural resources and sustainable development in engineering practices.	4	7
	CLO4			
	CLO5			

8. Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment :
Please select the Learning Outcome Domain (LOD) for each PLO in the cells above it. E.g. PLO1 - Knowledge, PLO2 - Cognitive, PLO3 - Practical Skills

Course Learning Outcomes (CLO)	**Programme Learning Outcomes (PLO)**												**Learning and Teaching Method**	**Assessment Method**
	Engineering Knowledge	Problem Analysis	Design/ Development of Solutions	Investigation	Modern Tool Usage	The Engineer and Society	Environmental and Sustainability	Ethics	Individual and Team Work	Communication	Project Management and Finance	Life Long Learning		
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**		
CLO1								√					Lecture, Tutorial	Assignment, Final Exam
CLO2						√							Lecture, Tutorial	Assignment, Final Exam
CLO3							√						Lecture, Tutorial	Assignment, Final Exam
CLO4														
CLO5														

Indicate the relevancy between the CLO and PLO by ticking "√" the appropriate relevant box.

(This description must be read together with Standards 2.1.2 , 2.2.1 and 2.2.2 in Area 2 - pages 16 & 18)

9.	**Transferable Skills (if applicable) (Skills learned in the course of study which can be useful and utilized in other settings)**		
		1	Leadership, autonomy and responsibility
		2	Ethics and professionalism
		3	
		4	

		5	

10. Distribution of Student Learning Time (SLT)

Course Content Outline	CLO*	Teaching and Learning Activities						SLT
		Guided Learning (F2F)				Guided Learning (NF2F) eg: e-Learning	Independent Learning (NF2F)	
		L	T	P	O			
1. Introduction to Ethics - Engineering Ethics - Personal vs Professional Ethics - Ethic and the Law	1	2	0				2	**4**
2. The Engineering Profession - Introduction to the Engineering Profession - Engineering Code of Ethics - Roles of Professional Bodies (BEM & IEM or/and IET & ECUK) - Route to Professional Engineer/Chartered Engineer - Case Studies	1	8	4				12	**24**
3. Technology and Society - Definition of technology and the role of technology in providing capabilities to society - The response and change of society to new capabilities - The development of new technologies to achieve desired functionality, socio-technological systems	2	4	2				6	**12**
4. Energy issues - Need for energy - Problems in the use of energy - Approaches to resolving problem of depletion and of environmental effect - Nuclear energy, hydrogen economy, etc.	3	6	3				9	**18**
5. Major environmental impact - Acid rain and global warming - Their impacts on society - Approaches in resolving the impacts	3	5	3				8	**16**
6. Risk , Safety and Accidents - Definition and determination of industrial/engineering safety and risks - Risk management and engineering safety analysis processes and tools - Theories of accidents - Legal considerations in industrial safety and accident preventions, safety and health legislation - Case studies of major industrial accidents and lessons learnt.	2	4	2				6	**12**
7. Sustainable Development Goals	3	1	1				2	**4**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
							0	**0**
Total		**30**	**15**	**0**	**0**	**0**	**45**	**90**

	Continuous Assessment		Percentage (%)	F2F	NF2F	SLT
1	Assignment		50		18	18
2						0
3						0
4						0
5						0
6						0
7						0
		Total	50	0	18	18

	Final Assessment		Percentage (%)	F2F	NF2F	SLT
1	Examination		50	2	10	12
2						0
3						0
4						0
5						0
		Total	50	2	10	12

****Please tick (√) if this course is Latihan Industri/ Clinical Placement/ Practicum/ WBL using Effective Learning Time(ELT) of 50%**

	GRAND TOTAL SLT	120

L = Lecture, T = Tutorial, P= Practical, O= Others, F2F=Face to Face, NF2F=Non Face to Face

**Indicate the CLO based on the CLO's numbering in Item 8.*

11	**Identify special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room, etc)**	None
12	**References (include required and further readings, and should be the most current)**	Required References • C. Fleddermann. 2014. Engineering Ethics. Pearson Education Limited. • Code of Professional Conduct, 2016, Board of Malaysis(BEM) • Occupational Safety and Health. Act 1994
13	**Other additional information :**	None